

Deep research: EMG datasets for separating voluntary movement from Parkinsonian tremor

Executive conclusion

The most important finding is unfortunately also the most useful one for planning the project:

I could not verify a publicly downloadable dataset that simultaneously satisfies all of the requirements:

Parkinson's disease or another pathological tremor + raw surface EMG from the hand/forearm + known electrode placement + simultaneous voluntary movement and tremor + synchronized kinematics + finger-level activity, preferably thumb/index finger.

There are published experiments that come **very close**. In particular, Widjaja et al. recorded **surface EMG together with accelerometry from Parkinsonian tremor** and explicitly developed a method for estimating voluntary motion in the presence of tremor. However, I could not verify a public raw-data repository associated with those experiments. ¹

Conversely, there are openly described/downloadable **hand sEMG datasets**, including the 2022 Özdemir et al. dataset with four-channel sEMG from 40 participants performing ten hand gestures, but these participants are not Parkinson's patients and the dataset therefore contains no genuine pathological tremor. ²

There is also an exceptionally relevant Parkinson's experiment by Ibrahim et al. involving **18 people with Parkinson's disease and sensors directly on the thumb/index-hand region**, including tasks in which subjects manipulated a pencil with the thumb and index finger. It is particularly relevant because the authors explicitly separate movement into a voluntary component and a tremor component. But the sensors are **IMUs, not EMG**, and I could not verify a public download of the patient recordings. ³

That means I would **not recommend building the project around the assumption that a ready-made Parkinson + finger-EMG dataset exists publicly**. A more realistic strategy is to combine:

1. a public hand-sEMG dataset to build and validate the EMG-processing code,
2. a Parkinson-tremor kinematic study/dataset to develop the voluntary-versus-tremor separation,
3. then either obtain the Parkinson EMG data from the original researchers or make a small purpose-built recording with EMG + IMU.

The third option may ultimately be much better scientifically because the electrode and sensor placement can be designed specifically for **thumb-index motion**, rather than trying to retrofit an unrelated dataset.

What the dataset search found

I searched combinations of *Parkinson*, *Parkinsonian tremor*, *essential tremor*, *pathological tremor*, *EMG*, *sEMG*, *electromyography*, *accelerometer*, *IMU*, *hand*, *finger*, *thumb*, *index finger*, *raw data*, and *dataset*, including targeted searches for PhysioNet, Zenodo, Figshare, Mendeley Data, Kaggle and general dataset indexing.

The result is a fairly clear mismatch between the literature and the public-data ecosystem: **many research groups have acquired exactly the kinds of EMG/accelerometer signals you want, but the raw recordings frequently were not deposited as public datasets.** The Widjaja Parkinson work is the clearest example. Meanwhile, modern public sEMG datasets are predominantly aimed at gesture recognition, prosthetic control or hand-pose estimation in participants without pathological tremor. ⁴

The following matrix summarizes the closest resources I could verify.

Dataset / study	Population	EMG	Motion sensor	Thumb/ index relevance	Voluntary + tremor	Public raw data verified?	Fit for your project
Widjaja et al. Parkinson tremor recordings	Parkinson's disease	Yes, sEMG	Yes, accelerometer	Upper limb; finger specificity unclear	Yes	No public repository verified	Excellent scientifically, poor accessibility ¹
Ibrahim et al. 2021 Parkinson recordings	18 people with PD	No	Yes, multiple IMUs	Excellent: thumb/ index directly involved	Yes	No public download verified	Excellent algorithmic benchmark, wrong sensor modality ⁵
Özdemir et al. multichannel sEMG hand dataset	40 participants, not pathological tremor	Yes, 4 channels	Not established as tremor ground truth	Hand gestures	No pathological tremor	Public dataset publication exists; file package should be checked before committing	Good for EMG pipeline development only ²

Dataset / study	Population	EMG	Motion sensor	Thumb/ index relevance	Voluntary + tremor	Public raw data verified?	Fit for your project
NinaPro DB2-type hand sEMG data	Non-Parkinson hand-movement subjects	Yes	Dataset-dependent	Strong hand/ finger relevance	No pathological tremor	Publicly used research database	Good for voluntary-movement EMG modelling, not tremor separation 6
Gallego et al. pathological-tremor recordings	Pathological tremor	No sEMG in the relevant method	Gyroscope	Upper limb	Yes	No reusable raw repository verified	Very useful separation algorithm reference 7

The important distinction is therefore between **a dataset suitable for developing individual parts of your algorithm** and **a dataset suitable for proving your final problem statement**. I found useful examples of the former; I did not find a publicly downloadable dataset that I would confidently claim satisfies the latter.

The most relevant datasets and experimental recordings

Widjaja et al.: Parkinsonian tremor with surface EMG and accelerometry

This is the **closest match to your actual problem formulation**.

Widjaja and colleagues published *Kalman Filtering of Accelerometer and Electromyography (EMG) Data in Pathological Tremor Sensing System* in 2008. The work combines accelerometer measurements with surface EMG specifically to estimate motion in pathological tremor. A related paper, *An Extended Kalman Filtering of Accelerometer and Surface Electromyography Data for Attenuation of Pathological Tremor*, describes the same general approach in more detail. ¹

The important point for your project is that they did not merely classify “tremor/no tremor.” Their objective is closer to yours: using EMG and inertial information to estimate the underlying limb motion while accounting for pathological tremor. The approach combines EMG-derived information, accelerometry, **Weighted-Frequency Fourier Linear Combiner (WFLC)** tremor-frequency tracking and an **Extended Kalman Filter (EKF)** to estimate components such as tremor amplitude, phase and bias. ⁸

Requirement	Assessment
Diagnosis	Parkinsonian/pathological tremor
EMG	Yes, surface EMG
Accelerometer	Yes

Requirement	Assessment
Raw signals	Used by investigators; public raw files not verified
Voluntary movement	Relevant to estimation framework
Pathological tremor	Yes
Sampling rate	Not sufficiently verified from the accessible indexed material to state confidently
Muscles/electrode positions	Upper-limb sEMG; I would verify the full paper before reproducing electrode placement
Thumb/index	No clear evidence of dedicated thumb/index channels in the material I could verify
Medication state	Not sufficiently documented in the accessible material for me to state
Public download	I could not verify one
Suitability	Very high if the authors' original data can be obtained

The biggest weakness is therefore not the experiment but **data access**. Scientifically, this is arguably the best literature match. Practically, it cannot currently be treated as an off-the-shelf public dataset.

My recommendation: this is one of the first author groups/papers I would cite in your problem analysis, because it establishes that combining sEMG with inertial measurements and model-based filtering is a sensible approach to pathological-tremor decomposition. ¹

Ibrahim et al.: Parkinson's disease with thumb/index IMUs

For the **thumb/index-finger part of your project**, this is the most interesting experiment I found.

Ibrahim et al.'s 2021 IEEE Transactions on Neural Systems and Rehabilitation Engineering paper, *Real-Time Voluntary Motion Prediction and Parkinson's Tremor Reduction Using Deep Neural Networks*, enrolled **18 people with Parkinson's disease: 11 men and 7 women, aged 60–84 years, mean 69 ± 7 years**. Sixteen of the 18 were taking Parkinson medication such as levodopa, pramipexole or amantadine, and the subjects were not asked to stop medication before the recordings. ⁹

The study is particularly valuable for you because sensors were placed on areas including the distal forearm/wrist and hand, with measurements associated with the **thumb and index/middle metacarpal regions and thumb/index MCP region**. Data were acquired at **100 Hz**. ⁹

Even more importantly, the protocol contained six tasks, including action tasks in which participants handled a pencil and performed movements involving the **thumb and index finger**. Rest, posture, wrist movement, pencil manipulation and spiral drawing were included, providing both tremor-dominant and voluntarily driven movement conditions. ¹⁰

Their baseline definition of the two components is directly relevant:

$$x(t) = v(t) + r(t)$$

where $v(t)$ represents voluntary movement and $r(t)$ tremor. For generating training targets, the authors treated voluntary movement primarily as a **0–2 Hz component** and Parkinsonian tremor as **3–17 Hz**, using fourth-order Butterworth filtering and forward-backward zero-phase filtering for the offline reference signals. ¹⁰

They subsequently trained a 1D-CNN/MLP framework. The paper reports approximately **99.2% performance for voluntary-motion estimation**, with future-motion prediction accuracy decreasing as the prediction horizon increased: roughly 97%, 94%, 91.6% and 89.9% for 10, 20, 50 and 100 future steps respectively; their system reported approximately 93.8% tremor reduction in the evaluated setup. ¹¹

But there is one enormous caveat:

This is an IMU dataset/experiment, not an EMG dataset.

I also could not verify an open raw-data repository associated with the paper.

Still, it is probably **the best experimental blueprint I found for your desired thumb/index protocol**. If you end up collecting your own EMG, you could reproduce several of these tasks almost directly while adding EMG to the muscles controlling the thumb and index finger.

Özdemir et al.: multichannel surface-EMG hand-movement dataset

Özdemir et al.'s 2022 Data in Brief publication provides a much more conventional **public hand-sEMG dataset**. The published description reports **four-channel time-series surface EMG from 40 participants performing ten distinct hand movements**. ²

This is highly useful for building software that handles:

- raw multichannel EMG,
- preprocessing,
- segmentation,
- feature extraction,
- gesture/movement classification,
- and eventually a model of the voluntary component.

However, it does **not contain Parkinsonian or essential tremor**, so it cannot validate a Parkinson-tremor separation algorithm. ²

Requirement	Assessment
Parkinson/ET	No
Participants	40
EMG	Yes, four-channel sEMG

Requirement	Assessment
Hand movements	Ten movement classes
Raw time series	Yes, dataset designed around time-series sEMG
Thumb/index	Hand movement relevant, but not specifically a Parkinson thumb-index tremor protocol
Tremor	No pathological tremor
IMU/ACC ground truth	Not established from the indexed dataset description I verified
Main value	Development/testing of your EMG processing software
Final validation	Unsuitable

This would therefore be a good **phase-one dataset**, but not the dataset on which I would make the final scientific claim that voluntary Parkinson movement can be separated from tremor.

NinaPro DB2 as a voluntary-movement reference

NinaPro is also worth considering as a **voluntary-hand-movement reference**, rather than a tremor dataset. Recent work uses NinaPro DB2 specifically for hand-movement recognition from sEMG; for example, Wu et al. used 300-ms windows and a set of 17 hand movements to train an LSTM/feature-imitation architecture. ¹² Other research likewise uses NinaPro datasets to evaluate hand-movement recognition algorithms from surface EMG. ¹³

The benefit compared with a generic EMG dataset is the breadth of hand activity. That makes NinaPro more interesting if you eventually want a model of:

$$\text{sEMG} \rightarrow \text{intended hand/finger movement}$$

before introducing pathological tremor.

Its limitation is fundamental: **no Parkinsonian tremor is present in the standard healthy-subject recordings**, so you cannot simply add a sinusoid to the EMG and claim to have reproduced pathological tremor. Pathological tremor is a change in neuromuscular activation and limb dynamics, not merely additive sensor noise.

I would therefore use NinaPro to answer questions such as “can our pipeline detect voluntary thumb/finger commands from forearm sEMG?” but **not** “can our system distinguish Parkinson tremor from voluntary movement?”

What the signal-processing literature suggests

The literature points toward three substantially different strategies. For your project, I would start with the first two before deep learning.

Frequency separation

The simplest hypothesis is:

$$x_{\text{movement}}(t) = x_{\text{voluntary}}(t) + x_{\text{tremor}}(t)$$

with the two components occupying sufficiently different frequency ranges to separate them approximately.

Ibrahim et al. explicitly operationalized that idea for Parkinsonian kinematics: voluntary movement was represented by frequencies below approximately **2 Hz**, whereas tremor was extracted over **3–17 Hz**. ¹⁰
Gallego et al. similarly developed real-time tremor estimation based on the premise that pathological tremor and voluntary movement have different frequency distributions. ⁷

That makes an FFT/PSD baseline entirely justified.

However, there is an important distinction for your EMG implementation:

Do not simply apply a 3–17 Hz band-pass to the raw EMG and call the result “tremor EMG.”

The frequency-domain separation in Ibrahim is performed on **movement/IMU information**, not on the raw myoelectric carrier. A more defensible EMG strategy is to preprocess the EMG, rectify it or derive an amplitude envelope, and then investigate rhythmic modulation of that envelope relative to the measured mechanical tremor. The synchronized accelerometer/gyroscope signal can identify the actual tremor frequency at each time point, after which corresponding periodic muscle activation can be sought in EMG.

That distinction is one reason why **EMG + IMU is dramatically preferable to EMG alone** for your problem.

Adaptive filtering and state estimation

The Widjaja work is especially relevant here. Rather than assuming a fixed tremor sinusoid, it employs **WFLC** to track tremor frequency and an **Extended Kalman Filter** to combine the information and estimate tremor parameters. ⁸

This matters because tremor is not perfectly stationary. A fixed band-pass filter can identify the overall tremor band, but an adaptive estimator can follow changes in:

$$f_{\text{tremor}}(t), \quad A_{\text{tremor}}(t), \quad \phi_{\text{tremor}}(t)$$

where frequency, amplitude and phase change over time.

Gallego et al. likewise investigated real-time estimation of pathological tremor from gyroscope signals and compared approaches intended to isolate involuntary tremor from the slower voluntary component. ⁷

For a bachelor/project-level implementation, **adaptive filtering or a Kalman-state model would be a particularly good second method to compare against simple Butterworth filtering**. It is sophisticated

enough to make the signal-processing project meaningful without immediately turning the whole problem into an opaque neural-network exercise.

Machine learning and deep learning

Ibrahim et al. demonstrate the next step: instead of prescribing the separation entirely through filters, a neural network learns a mapping between observed tremulous motion and an estimate of voluntary movement. Their 1D-CNN/MLP approach was trained using filtered voluntary-motion references generated from the recorded Parkinson data. ¹⁴

There is an important methodological lesson here: even a neural network still needs a definition of **ground truth**.

If your input is:

$$\text{EMG}_{\text{PD}}(t)$$

you normally do not possess a directly observable signal called:

$$\text{EMG}_{\text{voluntary,true}}(t).$$

Both voluntary commands and tremor-related activation occur in the patient's neuromuscular system. Therefore, simply training a neural network does not magically solve the labelling problem.

Ibrahim solved this for motion by deriving voluntary and tremor targets through frequency filtering of the kinematic recordings. ¹⁰ For your project, I would use the **IMU/position signal as the principal reference for mechanical voluntary movement and tremor**, while treating EMG as information about neural/muscular activation.

EMG decomposition

Motor-unit decomposition is a fourth, much more ambitious direction. Surface-EMG decomposition methods can attempt to recover individual motor-unit discharge activity from multichannel recordings rather than reducing the entire EMG channel to an RMS/envelope signal. De Luca et al. developed and evaluated methods for decomposing surface EMG into constituent motor-unit activity. ¹⁵

In principle, one could investigate whether particular motor-unit populations become strongly phase-locked to the tremor cycle while other activation relates to the desired movement.

For your first implementation, however, I would **not start here**. HD-sEMG decomposition brings its own substantial methodological problem and would make it much harder to determine whether failure is caused by the tremor-separation idea or by the motor-unit decomposition itself.

Recommended resources for your project

Because no single verified public dataset satisfies the whole problem, I would rank the resources according to **what they can actually contribute**, rather than pretending they are interchangeable.

Rank	Resource	Why I would use it	Main problem
Best scientific match	Widjaja Parkinson sEMG + accelerometer recordings	Correct pathology, EMG and mechanical tremor recorded together; algorithm closely matches your objective	Raw public download not verified ¹
Best thumb/index Parkinson match	Ibrahim et al. 2021	Actual PD patients, multiple sensors around hand/thumb/index, pencil/pinch-like tasks, explicit voluntary/tremor decomposition	IMU rather than EMG; no public raw dataset verified ⁵
Best accessible EMG starting point	Özdemir et al. hand-sEMG dataset	Real multichannel hand EMG; 40 people; ten movements; ideal for getting processing code working	Healthy/non-tremor participants ²
Best broader voluntary-hand benchmark	NinaPro DB2	Established sEMG hand-movement benchmark and useful for learning voluntary hand activity	No pathological tremor ⁶
Best algorithm reference without EMG	Gallego et al. pathological-tremor work	Directly studies real-time separation/estimation of tremor and voluntary movement	Gyroscope rather than EMG ⁷

Which is best for thumb and index finger?

There are actually two answers.

For Parkinson + thumb/index motion, Ibrahim et al. is by far the closest experimental match I found.

The hand instrumentation and tasks directly involve the thumb/index region, and the experiment was explicitly designed to separate Parkinson tremor from voluntary motion. ⁵

For Parkinson + EMG, Widjaja et al. is the closer match. It has the right disease and the right combination of sEMG and inertial sensing, but does not appear to give you the same digit-specific thumb/index protocol, and a public raw-data download was not found. ¹

Therefore, the **ideal dataset for your project would effectively be a fusion of those two experimental designs:**

Ibrahim thumb/index tasks + Widjaja EMG + IMU instrumentation

That would give you exactly what is currently missing from the public datasets I could verify.

Recommended first signal-processing pipeline

I would deliberately begin with a classical, interpretable pipeline rather than machine learning. It will give you a baseline and make it much easier to understand whether the signals actually contain enough information to solve the problem.

Signal model

Suppose an IMU on the finger gives measured motion

$$y(t) = v(t) + r(t) + n(t)$$

where:

- $v(t)$ = desired/voluntary motion,
- $r(t)$ = pathological tremor,
- $n(t)$ = sensor noise.

Meanwhile EMG from muscle i is

$$e_i(t)$$

and contains the muscle activation responsible for both intentional motion and tremor-related contractions.

The fundamental objective should therefore not initially be:

“Split raw EMG perfectly into voluntary EMG and tremor EMG.”

That is a much harder and somewhat ill-defined target.

A stronger first objective is:

Use EMG together with measured hand/finger kinematics to estimate which component of the resulting movement is voluntary and which is tremor.

That target has an observable output and can actually be evaluated.

EMG preprocessing

For each raw EMG channel I would start with the following chain:

$$e_{\text{raw}}(t) \rightarrow e_{\text{bandpass}}(t) \rightarrow e_{\text{notch}}(t) \rightarrow |e(t)| \rightarrow e_{\text{envelope}}(t)$$

A practical starting point would be a standard EMG band-pass appropriate to the hardware and sampling rate, power-line rejection where needed, full-wave rectification and a smooth envelope. The exact cut-offs should be chosen only after checking the acquisition system's sampling frequency and analogue filters rather than copied blindly across datasets.

Then normalize each channel per participant/session, for example using robust z-scores or task-specific normalization.

Find the actual tremor frequency from motion, not from assumptions

Use the accelerometer/gyroscope or finger-angle signal and estimate the power spectrum using Welch PSD over moving windows.

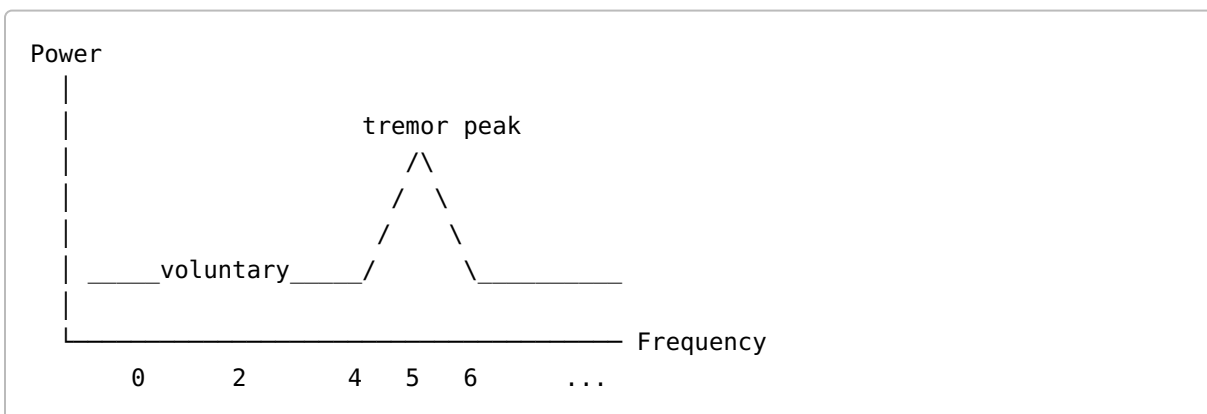
For every window, identify

$$f_T = \arg \max_f P_{\text{motion}}(f)$$

inside the plausible tremor band.

Ibrahim's 3–17 Hz tremor band and <2 Hz voluntary band give you a published starting baseline for Parkinsonian motion separation, but I would allow the dominant tremor frequency to be **patient-specific and time-varying** instead of fixing it. ¹⁰

Plotting the PSD should give a very intuitive first result:



This would immediately tell you whether the frequency-separation assumption is reasonable for each subject/task.

Build a simple kinematic baseline

Following the logic of Ibrahim et al., make the first reference estimate something like:

$$\hat{v}(t) = LPF_{2\text{ Hz}}\{y(t)\}$$

and

$$\hat{r}(t) = BPF_{\text{tremor}}\{y(t)\}.$$

Then verify:

$$y(t) \approx \hat{v}(t) + \hat{r}(t).$$

This is intentionally simple. The point is not for it to be your final algorithm; it gives you a target against which every more sophisticated algorithm can be evaluated. Ibrahim et al. used precisely this kind of spectral decomposition to construct voluntary and tremor references before training their network. 10

Relate tremor motion to EMG

Now calculate, for each muscle/channel:

- EMG-envelope PSD,
- motion/EMG cross-spectrum,
- magnitude-squared coherence,
- phase difference around f_T ,
- RMS/envelope amplitude around tremor episodes.

Conceptually:

$$C_{e_i,y}(f) = \frac{|S_{e_iy}(f)|^2}{S_{e_ie_i}(f)S_{yy}(f)}$$

and inspect

$$C_{e_i,y}(f_T).$$

If an EMG channel repeatedly shows strong rhythmic modulation coherent with the mechanical tremor frequency, it becomes a good candidate tremor-control signal.

This is much more informative than merely observing that “the EMG amplitude is high.”

Upgrade to WFLC plus Kalman filtering

Once the simple filter works, the next model I would implement is inspired directly by Widjaja et al.:

$$r(t) = A(t) \sin(2\pi f_T(t)t + \phi(t))$$

with A , f_T and ϕ allowed to change over time.

Use a WFLC/adaptive oscillator to track $f_T(t)$, then a Kalman or Extended Kalman Filter to estimate the tremor state and voluntary motion. That moves the project from:

fixed frequency filtering

to:

online adaptive separation of voluntary and involuntary motion.

This is very close to the signal-processing problem investigated by Widjaja et al. using accelerometry and sEMG in pathological tremor. ⁸

A sensible comparison for the report would therefore be:

Butterworth baseline

 vs WFLC + EKF

before introducing machine learning.

Evaluate whether voluntary movement was preserved

Do **not** evaluate the algorithm only by asking whether tremor energy decreases. An algorithm could achieve excellent “tremor suppression” by suppressing all movement.

I would evaluate at least two objectives:

$$\text{Tremor attenuation} = 10 \log_{10} \frac{P_T^{\text{before}}}{P_T^{\text{after}}}$$

and a voluntary-motion preservation metric such as correlation or RMSE:

$$RMSE_v = \sqrt{\frac{1}{N} \sum_{k=1}^N (v_{\text{ref}}[k] - \hat{v}[k])^2}.$$

The interesting engineering trade-off then becomes:

remove as much tremor as possible **without removing voluntary motion**

which is much closer to your actual problem statement than simply classifying Parkinson versus healthy subjects.

What I would record if you create your own small dataset

Given the lack of an ideal public dataset, a small targeted experiment could be unusually valuable.

I would record synchronized:

sEMG + thumb/index IMU + wrist/hand IMU

and use at least three categories of trials:

Trial	Purpose
Rest with hand supported	Estimate tremor without intentional movement
Postural hold	Characterize postural/action tremor

Trial	Purpose
Slow voluntary thumb/index task	Observe voluntary movement + tremor simultaneously
Repeated thumb-index pinch	Highly relevant functional movement
Pencil manipulation	Closely resembles Ibrahim's PD protocol
Tracking a slow reference trajectory	Gives an approximate known voluntary target

Ibrahim's protocol provides strong precedent for rest/postural conditions, wrist movement, pencil manipulation and spiral drawing in PD patients. ¹⁰

For your particular research question, the most valuable trial would be a **very slow controlled thumb-index movement while tremor remains present**. A visual target or metronome/trajectory can provide a reference for the intended low-frequency motion. That makes the separation problem considerably easier to validate.

For a future recording, I would also prioritize surface-accessible muscle sites relevant to the index/thumb movement and combine them with forearm flexor/extensor channels. The exact placement should be determined with someone competent in EMG anatomy, because finger-specific interpretation becomes difficult if all electrodes are placed generically around the forearm.

Bottom line

The research changes how I would formulate the project slightly.

I would **not** frame the initial technical problem as:

“Take Parkinson EMG and separate it into voluntary EMG and involuntary EMG.”

That implies that both components exist as clean, independently observable signals inside a single EMG channel, which makes validation difficult.

A much stronger formulation is:

“Investigate whether synchronized surface EMG and inertial measurements can be used to estimate and separate voluntary hand/finger motion from pathological Parkinsonian tremor, with particular focus on thumb-index movements.”

That formulation is directly supported by the closest Parkinson EMG/accelerometer work from Widjaja et al. and by the voluntary-motion/tremor decomposition and thumb/index protocol of Ibrahim et al. ¹⁶

For the practical project sequence, I would choose:

Stage A — build the EMG software: use the Özdemir hand-sEMG dataset or NinaPro to implement filtering, envelopes, PSD, segmentation and voluntary hand-movement features. ¹⁷

Stage B — build the separation algorithm: reproduce the simple low-frequency voluntary / tremor-band decomposition described by Ibrahim, then compare it with an adaptive WFLC/Kalman-style solution inspired by Widjaja and Gallego. ¹⁸

Stage C — validate on genuine Parkinson signals: the preferred target is synchronized **sEMG + IMU from a Parkinson patient performing a controlled thumb-index task**. Based on the public-data search, this is the main missing piece; the literature shows that such recordings have been made, but I could not verify an openly downloadable, finger-specific Parkinson sEMG dataset that provides it. ¹

So, among the resources found, **Widjaja is the best match to the signal-modality problem, Ibrahim is the best match to the Parkinson thumb/index movement problem, and Özdemir/NinaPro are the best immediate options for starting implementation with public sEMG**. None should be presented as the perfect dataset. That absence is itself an important research result—and it suggests that a modest, purpose-built Parkinson **EMG + IMU thumb/index dataset** could make your project considerably stronger than simply applying algorithms to an existing generic dataset.

¹ ⁴ <https://ieeexplore.ieee.org/document/4543706/>

<https://ieeexplore.ieee.org/document/4543706/>

² ¹⁷ <https://www.sciencedirect.com/science/article/pii/S2352340922001330>

<https://www.sciencedirect.com/science/article/pii/S2352340922001330>

³ ¹¹ ¹⁴ Real-Time Voluntary Motion Prediction and Parkinson's Tremor Reduction Using Deep Neural Networks - PubMed

<https://pubmed.ncbi.nlm.nih.gov/34255631/>

⁵ ⁹ ¹⁰ ¹⁸ https://www.researchgate.net/publication/353228763_Real-Time_Voluntary_Motion_Prediction_and_Parkinson%27s_Tremor_Reduction_Using_Deep_Neural_Networks

[https://www.researchgate.net/publication/353228763_Real-](https://www.researchgate.net/publication/353228763_Real-Time_Voluntary_Motion_Prediction_and_Parkinson%27s_Tremor_Reduction_Using_Deep_Neural_Networks)

[Time_Voluntary_Motion_Prediction_and_Parkinson%27s_Tremor_Reduction_Using_Deep_Neural_Networks](https://www.researchgate.net/publication/353228763_Real-Time_Voluntary_Motion_Prediction_and_Parkinson%27s_Tremor_Reduction_Using_Deep_Neural_Networks)

⁶ ¹² <https://arxiv.org/abs/2405.19356>

<https://arxiv.org/abs/2405.19356>

⁷ <https://www.mdpi.com/1424-8220/10/3/2129>

<https://www.mdpi.com/1424-8220/10/3/2129>

⁸ ¹⁶ [https://www.researchgate.net/publication/](https://www.researchgate.net/publication/224376066_An_Extended_Kalman_Filtering_of_Accelerometer_and_Surface_Electromyography_Data_for_Attenuation_of_Pathological_Tremor)

[224376066_An_Extended_Kalman_Filtering_of_Accelerometer_and_Surface_Electromyography_Data_for_Attenuation_of_Pathol](https://www.researchgate.net/publication/224376066_An_Extended_Kalman_Filtering_of_Accelerometer_and_Surface_Electromyography_Data_for_Attenuation_of_Pathological_Tremor)

[https://www.researchgate.net/publication/](https://www.researchgate.net/publication/224376066_An_Extended_Kalman_Filtering_of_Accelerometer_and_Surface_Electromyography_Data_for_Attenuation_of_Pathological_Tremor)

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¹³ <https://www.sciencedirect.com/science/article/pii/S0208521621000346>

<https://www.sciencedirect.com/science/article/pii/S0208521621000346>

¹⁵ <https://pubmed.ncbi.nlm.nih.gov/16899649/>

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